Urban Policies to Combat Climate Change: Assessing the Impact of Congestion Pricing and Low-Emission Zones using Machine Learning Techniques

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Executive Summary

Abstract content goes here. This section should provide a concise summary of the research question, methodology, key findings, and implications of the study. It should be clear and informative, allowing readers to quickly understand the purpose and results of the report.

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Replication files are available in the GitHub repository for this project.

1 Introduction

Discussions about global warming have been highly prevalent in public discourse, especially in the past 10 to 15 years. Many policies have been implemented, and even more have been proposed and advocated. Some are to be decided on the supranational level, e.g., the EU's climate certificate trading, some on the national level, e.g., electric vehicle subsidies, but many can also be implemented on the regional or even municipal level. Especially when it comes to reducing transport CO_2 emissions, many decisions fall under the authority of the municipal government, be it how road infrastructure is designed, what speed limits are set, or further driving regulations. This report estimates the effectiveness in reducing transport carbon emissions of two such regulatory policies:

Definition 1. ***Congestion Pricing (CP):*** *Surcharges are implemented in certain areas of a city for cars that enter from outside.*

Definition 2. ***Low-Emission Zones (LEZ):*** *Certain vehicles (older diesel or petrol engines) are banned in certain areas of a city.*

I study these two policies in a large panel dataset of many European cities, focusing especially on three core research questions:

1. Were the policies effective in reducing emissions?
2. Which type of cities benefit the most/least from the policies?
3. Does implementation of both policies simultaneously produce greater emission reductions than what would have been expected from the sum of the individual effects of implementing either policy alone?

My findings show that ....

The remainder of the report is structured as follows. Section 2 introduces the empirical strategy and the identification assumptions underlying the analysis. Section 3 outlines the proposed estimation framework, while Section 4 discusses its theoretical properties and the relevance of orthogonality. Section 5 describes the construction of the empirical dataset and the clustering procedure used to distinguish city types. Section 6 presents the main empirical findings, Section 7 evaluates robustness and sensitivity checks, and Section 8 provides an interpretation of the results and their policy implications.

2 Empirical Strategy and Identification

The target parameters of this study are the Average Treatment Effect on the Treated (ATT) and the Group Average Treatment Effect on the Treated (GATT).

The global ATT measures the average impact of a specific policy regime across all cities that actually implemented it, evaluated against the baseline of no policy ($d = 0$). It is defined as:

$$\tau_{ATT,d} = \mathbb{E}[Y_{it}(d) - Y_{it}(0) \mid D_{it} = d] \tag{1}$$

In this setting, treatment is not binary but has three states: CP ($d = 1$), LEZ ($d = 2$), and their synergy CP $\times$ LEZ ($d = 3$).

To address research question 2 regarding spatial heterogeneity, the GATT isolates this treatment effect within the specific structural typologies G_i. The two clusters $G_i \in \{0, 1\}$ are constructed using k-means clustering on $X_i = [X_i^{invariant}, \bar{X}_i^{pre-treatment}]$, both time-invariant factors like geographical features as well as pre-treatment averages of covariates, so called Mundlak devices (see Section 5). The GATT for each cluster is therefore defined as:

$$\tau_{GATT,d}(g) = \mathbb{E}[Y_{it}(d) - Y_{it}(0) \mid D_{it} = d, G_i = g] \quad \text{for } g \in \{0, 1\} \tag{2}$$

For the ATT and GATT to be identified correctly, three assumptions are required.

Assumption 1. Conditional Independence/Ignorability: $(Y(d), Y(0)) \perp\!\!\!\perp D \mid X_i$. Conditional on the Mundlak device and baseline geography, treatment assignment is independent of potential outcomes. I.e., variables of X_i encompass all relevant factors that determine whether a city implements policy d (i. Chernozhukov et al., 2024, Ch. 5).

Assumption 2. Weak Overlap: $P(D = 1 \mid X) < 1$. Policy implementation cannot be perfectly predicted by the pre-treatment covariates X_i. I.e., every treated city has a similar untreated counterpart (i. Chernozhukov et al., 2024, Ch. 5).

Assumption 3. SUTVA: The potential outcomes of city i are unaffected by the treatment status of city j (no spatial spillovers) (i. Chernozhukov et al., 2024, Ch. 2).

3 Proposed Estimator

To estimate the (G)ATT, I employ two different double machine learning approaches: Augmented Inverse Propensity Weighting (AIPW) and Causal Forest, estimating the treatment effects in two stages:

Stage 1 (Orthogonalization): The algorithm first uses LASSO to estimate two "nuisance" functions using cross-fitting with 5 folds, whereby the hyperparameters α and C are determined once outside the cross-fitting to save computation time. The two nuisance functions are as such:

- The propensity score: $p(W_{it}) = \mathbb{E}[D_{it} \mid W_{it}]$ is estimated via a logistic regression with L1 Lasso penalty. Then the Treatment Residual: $\tilde{D}_{it} = D_{it} - \hat{p}(W_{it})$ is calculated as the leftover residual of the true treatment D_{it} minus the predicted propensity score $\hat{p}(W_{it})$ for each city-time observation, representing how much of the variable could not be explained by the regression.

- The expected outcome: $m(W_{it}) = \mathbb{E}[Y_{it} \mid W_{it}]$ is estimated via a linear LASSO regression. Then the Outcome Residual: $\tilde{Y}_{it} = Y_{it} - \hat{m}(W_{it})$ is calculated as the leftover residual of the true outcome Y_{it} minus the predicted outcome $\hat{m}(W_{it})$ for each city-time observation.

Both of these LASSO regressions are fit with the whole set $W_{it} = X_{it} + X_i$, both contemporary and time-invariant covariates, whereby the LASSO penalty shrinks down and eliminates covariates to minimize prediction error in a cross-validation setting.[1] This stage is the same for both AIPW and Causal Forest (See V. Chernozhukov et al., 2018).

[1] W_it also omits other variables that are expected to be impacted by or to impact either of the two policies such as e.g. PM2.5 emissions, vehicle engine shares, or records of public announcement of such policies.

Stage 2 (ATT/GATT Estimation): Here AIPW and Causal Forest differ in how they use the DR scores to estimate the treatment effects.

AIPW uses the treatment and outcome residuals ($\tilde{D}_{it}$ and $\tilde{Y}_{it}$) to estimate Doubly Robust scores (DR), a pseudo-outcome of how a specific city-year observation should have been based on what the first stage models can explain. The GATT then represents the average of all individual DR scores of treated cities within a cluster, and the ATT the average of all treated cities (See V. Chernozhukov et al., 2018).

Causal Forest builds a large set of trees, 200, whereby each tree uses a random subsample of cities. Each tree is very shallow, consisting of only two terminal nodes (leaves), the clusters for dense metropolises and regional hubs. Then, within each leaf, an OLS regression is fitted on the orthogonalized residuals from Stage 1 ($\tilde{Y}_{it}$ and $\tilde{D}_{it}$). Finally, the estimations of all different leaves are averaged for the final GATT estimate (and then the two GATT estimates averaged for the ATT estimate). The shallowness of the tree is beneficial as it makes estimation within each leaf well estimable, as each leaf still contains a large number of treated and untreated cities, which is especially necessary for estimating the synergy effect, which was relatively rarely implemented in the dataset (See Wager and Athey, 2018; Athey et al., 2019).

4 Theoretical Properties

Both methods rely on Neyman orthogonality as proof for their unbiasedness. Let us first recall the general theorem of Neyman orthogonality.

Theorem 1. ***Neyman Orthogonality:*** *A score function $\psi(Z;\theta,\eta)$ identifying a causal parameter θ with nuisance parameters η is Neyman orthogonal if the derivative of the expected score with respect to the nuisance parameters, evaluated at their true values η_0, equals zero:*

$$\partial_\eta \mathbb{E}[\psi(Z;\theta_0,\eta_0)] = 0 \tag{3}$$

Both AIPW and Causal Forest circumvent the inherent bias of Lasso estimation (which stems from the penalization) by their specification of the second stage, thereby eliminating transmission of bias from the first stage Lasso regularized estimation into the final estimates. Their confirmation of Neyman orthogonality is shown in the following two proofs.

Theorem 2. ***Neyman Orthogonality:*** *Let $Z = (Y, D, W)$ denote the full data vector. A score function $\psi(Z;\theta,\eta)$ identifying a causal parameter θ with nuisance parameters η is Neyman orthogonal if the Gateaux derivative of the expected score with respect to the nuisance parameters, evaluated at their true values η_0, equals zero:*

$$\partial_\eta \mathbb{E}[\psi(Z;\theta_0,\eta_0)] = 0 \tag{4}$$

Both AIPW and the Causal Forest circumvent the inherent regularization bias of Lasso (which stems from L1 penalization) by specifying orthogonal second-stage moments. This prevents the transmission of first-stage prediction errors into the final causal estimates. Their confirmation of Neyman orthogonality is demonstrated in the following two proofs.

Theorem 3. ***Neyman Orthogonality:*** *Let $Z = (Y, D, W)$ denote the full data vector. A score function $\psi(Z;\theta,\eta)$ identifying a causal parameter θ with nuisance parameters*

η is Neyman orthogonal if the pathwise (Gateaux) derivative of the expected score with respect to the nuisance parameters, evaluated at their true values η_0, equals zero for all permissible perturbation directions h:

$$\partial_r \mathbb{E}[\psi(Z;\theta_0,\eta_0+rh)]\Big|_{r=0}=0$$

Both AIPW and the Causal Forest circumvent the inherent regularization bias of Lasso (which stems from L1 penalization) by specifying orthogonal second-stage moments. This prevents the transmission of first-stage functional prediction errors into the final causal estimates. Their confirmation of Neyman orthogonality is demonstrated in the following two proofs.

Proof. **AIPW (Doubly Robust Score for the Treated):** Let the parameter of interest θ be the expected potential outcome under treatment $\mathbb{E}[Y(1)]$. The true nuisance functions are $\eta_0=(\mu_{1,0},p_0)$, where $\mu_{1,0}(W)=\mathbb{E}[Y\mid D=1,W]$ and $p_0(W)=\mathbb{E}[D\mid W]$.

The score function is defined as:

$$\psi(Z;\theta,\eta)=\mu_1(W)+\frac{D}{p(W)}\Big(Y-\mu_1(W)\Big)-\theta$$

Orthogonality with respect to the outcome model $\mu_1(W)$: We consider a perturbed function $\mu_{1,r}(W)=\mu_{1,0}(W)+rh_\mu(W)$ and differentiate with respect to r at $r=0$. Applying the Law of Iterated Expectations (LIE) conditioning on W:

$$\frac{\partial}{\partial r}\mathbb{E}\left[\mu_{1,0}(W)+rh_\mu(W)+\frac{D}{p_0(W)}\Big(Y-\mu_{1,0}(W)-rh_\mu(W)\Big)-\theta_0\right]\Bigg|_{r=0}$$

$$=\mathbb{E}\left[h_\mu(W)-\frac{D}{p_0(W)}h_\mu(W)\right]=\mathbb{E}_W\left[h_\mu(W)\left(1-\frac{\mathbb{E}[D\mid W]}{p_0(W)}\right)\right]=\mathbb{E}_W\left[h_\mu(W)(1-1)\right]=0$$

Orthogonality with respect to the propensity model $p(W)$: We consider a perturbed function $p_r(W)=p_0(W)+rh_p(W)$ and differentiate with respect to r at $r=0$. Again, applying the LIE:

$$\frac{\partial}{\partial r}\mathbb{E}\left[\mu_{1,0}(W)+\frac{D}{p_0(W)+rh_p(W)}\Big(Y-\mu_{1,0}(W)\Big)-\theta_0\right]\Bigg|_{r=0}$$

$$=\mathbb{E}\left[-\frac{D\cdot h_p(W)}{p_0(W)^2}\Big(Y-\mu_{1,0}(W)\Big)\right]=\mathbb{E}_W\left[-\frac{h_p(W)}{p_0(W)^2}\underbrace{\mathbb{E}[D(Y-\mu_{1,0}(W))\mid W]}_{=0}\right]=0$$

□

Proof. **Causal Forest (Robinson Transformation / R-Learner):** Let the parameter θ be the average treatment effect. The true nuisance functions are $\eta_0=(m_0,p_0)$, where $m_0(W)=\mathbb{E}[Y\mid W]$ and $p_0(W)=\mathbb{E}[D\mid W]$.

The score function for the partialled-out regression is:

$$\psi(Z;\theta,\eta) = \Big(Y - m(W) - \theta(D - p(W))\Big)\Big(D - p(W)\Big)$$

Orthogonality with respect to the marginal outcome model $m(W)$: Consider the perturbation $m_r(W) = m_0(W) + rh_m(W)$. Evaluating the derivative at $r = 0$ and applying the LIE:

$$\frac{\partial}{\partial r}\mathbb{E}\left[\Big(Y - (m_0(W) + rh_m(W)) - \theta_0(D - p_0(W))\Big)\Big(D - p_0(W)\Big)\right]\Bigg|_{r=0}$$

$$= \mathbb{E}\left[-h_m(W)(D - p_0(W))\right] = \mathbb{E}_W\left[-h_m(W)\underbrace{\mathbb{E}[D - p_0(W) \mid W]}_{=0}\right] = 0$$

Orthogonality with respect to the propensity model $p(W)$: Consider the perturbation $p_r(W) = p_0(W) + rh_p(W)$. Applying the product rule for the derivative with respect to r at $r = 0$:

$$\frac{\partial}{\partial r}\mathbb{E}\left[\Big(Y - m_0(W) - \theta_0(D - (p_0(W) + rh_p(W)))\Big)\Big(D - (p_0(W) + rh_p(W))\Big)\right]\Bigg|_{r=0}$$

$$= \mathbb{E}\left[\Big(\theta_0 h_p(W)\Big)\Big(D - p_0(W)\Big) + \Big(Y - m_0(W) - \theta_0(D - p_0(W))\Big)\Big(-h_p(W)\Big)\right]$$

$$= \mathbb{E}\left[h_p(W)\Big(2\theta_0(D - p_0(W)) - (Y - m_0(W))\Big)\right]$$

$$= \mathbb{E}_W\left[h_p(W)\Big(2\theta_0\underbrace{\mathbb{E}[D - p_0(W) \mid W]}_{=0} - \underbrace{\mathbb{E}[Y - m_0(W) \mid W]}_{=0}\Big)\right] = 0$$

□

5 Data Preparation

The raw panel dataset, consisting of 51 variables for 183 cities in the time frame from 2008 to 2024, has to be adjusted for a valid estimation via double machine learning. First, minor adjustments are made by changing the categorical variable on the latitude zone into separate dummies as well as log-transforming variables that have highly skewed distributions and/or where it is economically meaningful to do so.

For accurate estimation in a panel setting, time-invariant traits have to be controlled for. For this purpose, the dataset already includes some time-invariant covariates $X_i^{geographic}$ depicting geographic features. Additionally, I transform all time-, non-dummy features into time-invariant Mundlak devices $\bar{X}_i^{pre}$ by taking the within-city average over all periods before the implementation of one of the two policies (See Mundlak, 1978). This limitation on pre-periods ensures that potential side-effects of the policies are not altering the averages. One variable that didn't change for any city in any pre-period is dropped: a dummy for signing a national climate pact. Additonally, I add year dummies in order to account for time fixed effects.

K-means clustering

In the next step, I utilize the whole set of time-invariant covariates X_i with the exception of latitude dummies in order to run k-means clustering to determine a good split of the city features into two types (See James et al., 2023, Ch. 12.4.1). The benefit of using clusters instead of raw covariates for the second stage of the estimations is *1.* that it ensures that the split for the shallow tree of the causal forest is explainable and *2.* to answer research question 2 not just based on single covariate differences but an overarching typology of city structure. The number of $K = 2$ clusters is chosen based on the criterion of explainability as well as backed by silhouette analysis (see Figure 1). For the cluster determination, I standardize the X_i variables to have mean 0 and a standard deviation of 1 in order to make their magnitude comparable across different metrics. The optimal clusters are found by Euclidean partitioning. First, $K = 2$ centroids, i.e., specifications of all the different variables, are randomly chosen, and then the distance across all dimensions for each city to those centroids is calculated, and each city is assigned to the closer centroid. Then the centroid is adjusted to the mean for all variables that are assigned to it. Now the distances to this adjusted centroid are calculated for each city, and the process iterates forward like this until there is no further centroid adjustment.

Then finally, after inspecting the structural drivers amongst all the factors X_i for each cluster, the typology of both groups becomes clear. *Cluster 0* consists of area-wise large cities with smaller population, more petrol engine vehicles, and that has a large logistics share. *Cluster 1* consists of dense, and population-wise large cities that have better public transit, more cars with Diesel and electric engines than diesel engines, and a higher share of green party votes in elections (also see Figure 2 in the Appendix). Based on these criteria, I name *Cluster 0* as "Sprawling Regional Hub" and *Cluster 1* as "Dense Metropolis".

Table 1: Distribution of Climate Policy Treatments by Urban Typology

Urban Typology	Never Treated	CP Only	LEZ Only	Synergy (CP × LEZ)	Total
Dense Metropolises (1)	11	27	16	25	79
Regional Hubs (0)	62	13	26	3	104
Total	**73**	**40**	**42**	**28**	**183**

Notes: Table displays the count of unique cities. A city is classified as “Ever Treated” if Congestion Pricing was active in any year during the sample period.

Table 1 shows that these clusters both have a sufficient number of cities for both single policies, Congestion Pricing and Low-Emission Zones; however, the cities that implemented both are strongly concentrated in *Cluster 1*. Hence, synergy effects cannot reliably be estimated for separate city types.

6 Results

For all the estimations, AIPW and Causal forest find relatively similar point estimates, indicating that, if the assumptions hold, both methods come to estimates that are mostly very similar in coefficient size and agree in statistical significance. I will first answer the three research questions based on the main results in Table 2 and then discuss possible concerns surrounding the magnitude of the estimates and their significance in Section 7.

Table 2: Estimates of Policy Effect on Y = *Transport CO_2 emissions*

Policy	(1) AIPW	(2) Causal Forest
Panel A: Average Treatment Effect on the Treated (ATT)		
Congestion Pricing (CP)	-0.3304^{**} ($p = 0.016$)	-0.3837^{***} ($p = 0.000$)
Low Emission Zone (LEZ)	-0.1111^{**} ($p = 0.042$)	-0.2014^{**} ($p = 0.014$)
Synergy (CP $\times$ LEZ)	-1.3174^{***} ($p = 0.000$)	-1.7564^{***} ($p = 0.000$)
Panel B: Group ATT (Heterogeneity by City Type)		
Cluster 1: Dense Metropolis		
Congestion Pricing (CP)	-0.2010 ($p = 0.172$)	-0.2190 ($p = 0.519$)
Low Emission Zone (LEZ)	-0.3758^{***} ($p = 0.007$)	-0.6839^{***} ($p = 0.000$)
Cluster 0: Sprawling Regional Hub		
Congestion Pricing (CP)	-0.3407^{**} ($p = 0.017$)	-0.3968^{***} ($p = 0.000$)
Low Emission Zone (LEZ)	-0.0817 ($p = 0.177$)	-0.1479^{*} ($p = 0.098$)

Notes: p-values in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. 'Failed' estimation bounds denote overlap violations.

Research Question 1. Congestion pricing as well as low-emission zones are estimated to have a negative effect on transport CO_2 emissions. Congestion Pricing is estimated to reduce emissions by 28% (AIPW) to 32% (Causal Forest). Low-Emission Zones are expected to lead to a reduction of 10% (AIPW) to 18% (Causal Forest). The coefficients for both policies are significant to the 5 % level under both estimation procedures.

Research Question 2. Panel B in Table 2 shows separate estimations for the GATT for both clusters. Interestingly, it is found that Low-Emission Zones only have a significant effect for cities of *Cluster 1*, the Dense Metropolis, whereas Congestion Pricing only exhibits a significant effect to the 5% level for Sprawling Regional Hubs in *Cluster 0*. It is estimated that LEZs are highly effective in the Dense Metropolis type, reducing transport emissions by 32% (AIPW) to nearly 50% (Causal Forest). Similarily, CP is shown to be more effective for cities in *Cluster 0* than in the general ATT; with an estimated reduction of around 30% for both AIPW and Causal Forest.

Research Question 3. The most startling result is definitely the coefficient for the synergy effects of implementing both CP and a LEZ simulataneously. Both AIPW and Causal Forest estimate that this combination of policies lead to a reduction of CO_2 in treated cities by at least 73% (AIPW) or even staggering 83% (Causal Forest). This is a lot more than the simple addition of both policy's effects which is just 36% (AIPW) or 44% (Causal Forest). Therefore, the additional synergy effect can be determined to be around 40% for both methods.

7 Sensitivity Analysis

To investigate whether the results are robust estimations of the true causal effect of policy implementation, it is important to check whether the comparison of treated and untreated cities is valid. For this, one can look at the distribution of propensity scores to determine whether the implementation of the policy is highly determenistic or whether many untreated observations are also predicted to have a large probability of implementing a policy. Table 3 shows the common support region and retainment rates, i.e. the range of propensity scores that are existent in both treatment and control group and how many observations had to be excluded because they are outside of this range. Congestion Pricing has the best common support as more than 90% of observations with CP implementation and more than 75% untreated observations fall within the region showing that the policy is not too strongly predicted as well as keeping a large enough sample for estimation. Nevertheless, even for this policy regime, the distribution of propensity scores diverges strongly between the two groups (See Table 4 and Figure 4 in the Appendix). Low-Emission Zones also still have decent coverage with nearly all treated observations and more than half of the untreated sample being kept for estimation. The overlap for both is not optimal but decent enough for the estimation. However, for the synergy of both CP and LEZ common support is strictly violated. As a rare occurrence with in total only 28 cities, the propensity score distribution reveals that there is strong selection into this policy, making it strongly determined by previous city-specific factors. While the propensity score for treated observations is already 19% in the minimum and even 68% in just the 25th percentile, untreated units are strongly concentrated at propensity scores of 0. Therefore, only 3.6% of untreated observations can even be retained in the estimation. This constitutes a violation of Assumption 2 (Weak Overlap), thereby rendering the results unreliable. Inference of possible synergetic effects is therefore invalid. On the other hand, the overlap for Congestion Pricing and Low-Emission Zones is solid enough as the coefficients also remain stable when increasing the trimming in AIPW, i.e. the exclusion of observations with propensity scores to close to 1 or 0 in order to protect the weighting. Across trimming rates of 0.01 (baseline result value), 0.05, and 0.1, the coefficients remain largely the same; albeit a small reduction in the coefficient for LEZ renders it just outside the bounds of the 5% significance level (See Figure 5).

As an additional sensitivity test, I conduct placebo estimations for the variables "Number of Libraries" and "Streetlight Density", two factors that should be unrelated to any of the two policies. Table 5 (in the Appendix) confirms that all coefficients are estimated to be statistically insignificant, giving a good indication that AIPW and Causal Forest do not inhabit hidden flaws.

Table 3: Common Support Bounds and Sample Retention by Policy

Policy	Support Region	Treated Retained	Untreated Retained
Congestion Pricing (CP Only)	$[0.004, 0.794]$	346 (91.1%)	2063 (75.5%)
Low Emission Zone (LEZ Only)	$[0.016, 0.914]$	305 (98.1%)	1522 (54.4%)
Synergy (CP $\times$ LEZ)	$[0.186, 0.940]$	255 (90.7%)	102 (3.6%)

Notes: The common support region is defined mathematically as $[\max(\min_T, \min_C), \min(\max_T, \max_C)]$ for each respective policy's propensity score distribution. Units outside these bounds are strictly trimmed prior to causal estimation to ensure finite-sample overlap.

Table 4: Propensity Score Distributions Across All Policy Regimes

	Congestion Pricing		Low Emission Zone		Synergy (CP × LEZ)	
Statistic	Treated	Untreated	Treated	Untreated	Treated	Untreated
Observations (N)	380	2731	311	2800	281	2830
Mean	0.408	0.074	0.375	0.071	0.762	0.021
Std. Dev.	0.253	0.123	0.246	0.122	0.142	0.081
Minimum	0.004	0.000	0.016	0.000	0.186	0.000
25th Pctl.	0.186	0.004	0.163	0.004	0.680	0.000
Median	0.367	0.022	0.335	0.021	0.778	0.001
75th Pctl.	0.605	0.083	0.558	0.085	0.870	0.006
Maximum	0.921	0.794	0.974	0.914	0.994	0.940

Notes: Table displays the descriptive statistics of the estimated propensity scores prior to overlap trimming. "Untreated" refers strictly to baseline ($D = 0$) units.

Table 5: Falsification Tests: Estimated Effects on Placebo Outcomes (Global ATT)

Decoy Outcome	Congestion Pricing (CP)	Low Emission Zone (LEZ)	Synergy (CP × LEZ)
Panel A: AIPW (Lasso)			
No. of Libraries	-0.0381 ($p = 0.302$)	0.0522 ($p = 0.404$)	0.1275 ($p = 0.495$)
Streetlight Density	0.8928 ($p = 0.407$)	-0.8942 ($p = 0.309$)	-0.4635 ($p = 0.696$)
No. of Fountains	-0.2429 ($p = 0.479$)	-0.0715 ($p = 0.663$)	-0.1922 ($p = 0.712$)
No. of benches pc	0.0317 ($p = 0.799$)	-0.4620^{***} ($p = 0.000$)	-0.1622 ($p = 0.201$)
Panel B: Causal Forest			
No. of Libraries	-0.0724^{**} ($p = 0.024$)	0.1015^{**} ($p = 0.026$)	0.0039 ($p = 0.966$)
Streetlight Density	1.7836^{**} ($p = 0.020$)	0.0465 ($p = 0.952$)	0.8528 ($p = 0.390$)
No. of Fountains	-0.4861 ($p = 0.133$)	-0.2782 ($p = 0.295$)	-0.2719 ($p = 0.546$)
No. of benches pc	-0.0180 ($p = 0.878$)	-0.5141^{***} ($p = 0.000$)	-0.3554^{***} ($p = 0.004$)

Notes: p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

8 Interpretation and Discussion

- Some evidence that policies may have been effective but results to be interpreted with caution.

- Interpretation of GATT difference: This result may seem reasonable; cities in *Cluster 1* offer more substitutions that become attractive after CP or an LEZ is implemented. Commuters can switch to higher quality public transit, or are more willing to switch to electric vehicles as the share is already larger in the city. As the logistics industry also has a lesser role in these cities, there is less logistics transit that would still have to continue (albeit possibly with routes outside the proclaimed zones). However, also here the significance of the results is debatable.

- Point out methods benefits but that the results may be influenced by the issues

The approach chosen in this study, DML with AIPW and Causal Forest estimation, relies on the assumption that conditioning on the large set W_it of contemporaneous controls, Mundlak devices as well as time fixed effects the decision to implement a policy or not is effectively random. The sensitivity analysis showed that this is partly given, yet the treated and untreated observations still retain structural differences. An alternative to circumvent this assumption would be to conduct a Staggered Difference-in-Differences (DiD) event study following Callaway and Sant'Anna (2021). Yet this approach then requires the parallel trends assumption, i.e. the outcome variable, transport CO_2 emissions, exhibits the same time trend for both treated and untreated cities in the absence of treatment. If you, the Board of Mayors, wish to get an additional robustness check on the efficacy of Congestion Pricing and Low-Emission Zones, commissioning a second study using the Callaway and Sant'Anna (2021) Doubly-Robust DiD approach may be a worthwhile endeavour.

- contrast with alternative of staggered DiD

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Appendix

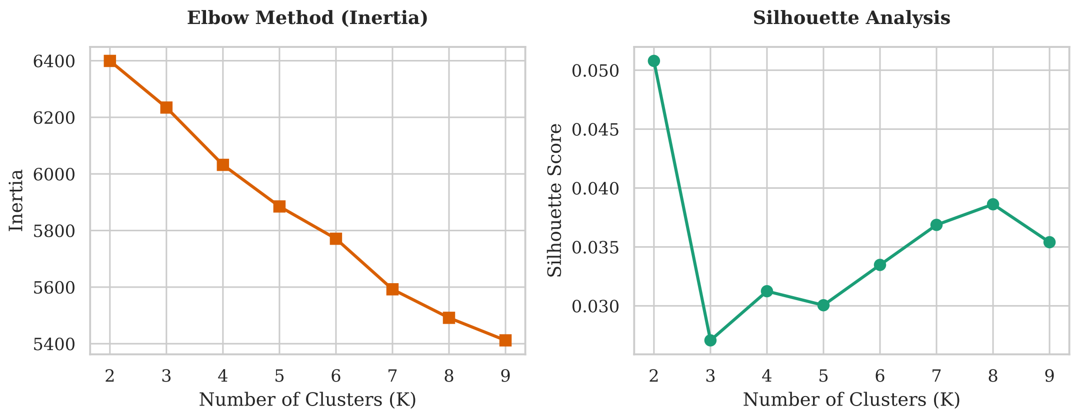

Figure 1: Evaluation of k-Means Number of Clusters.

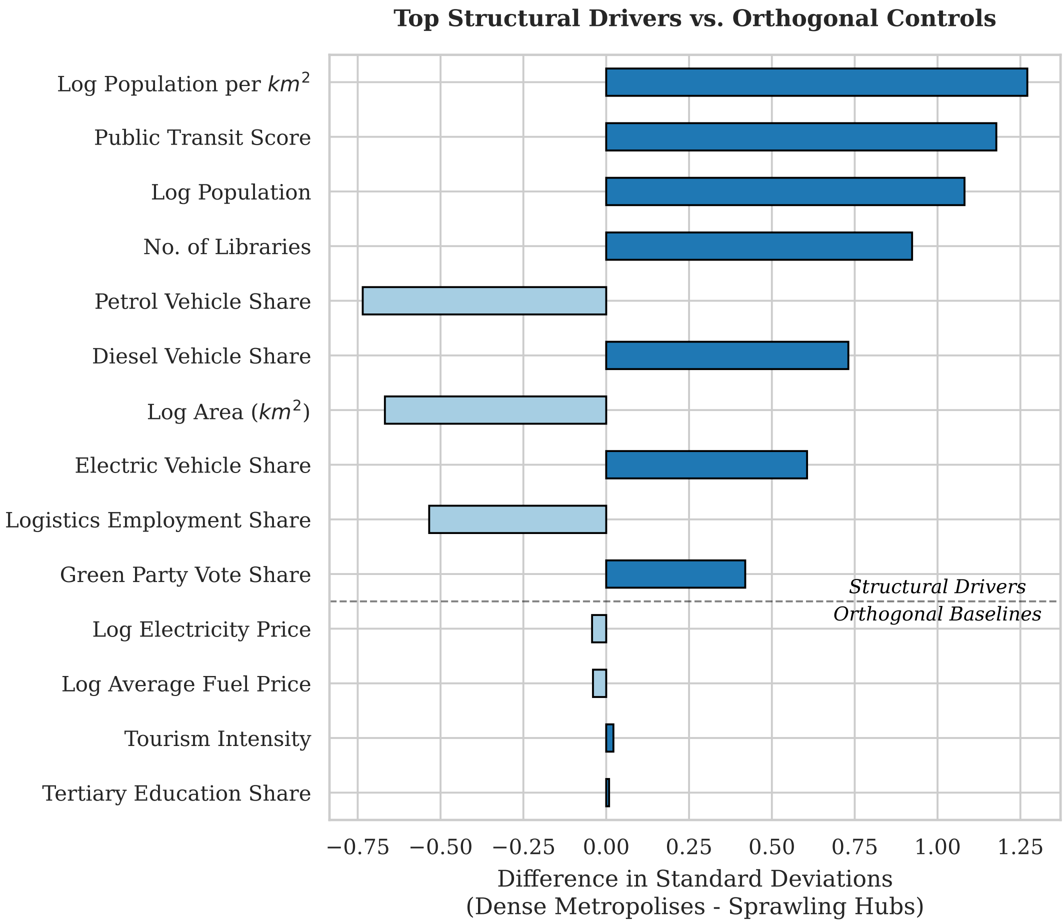

Figure 2: Differences between the two clusters.

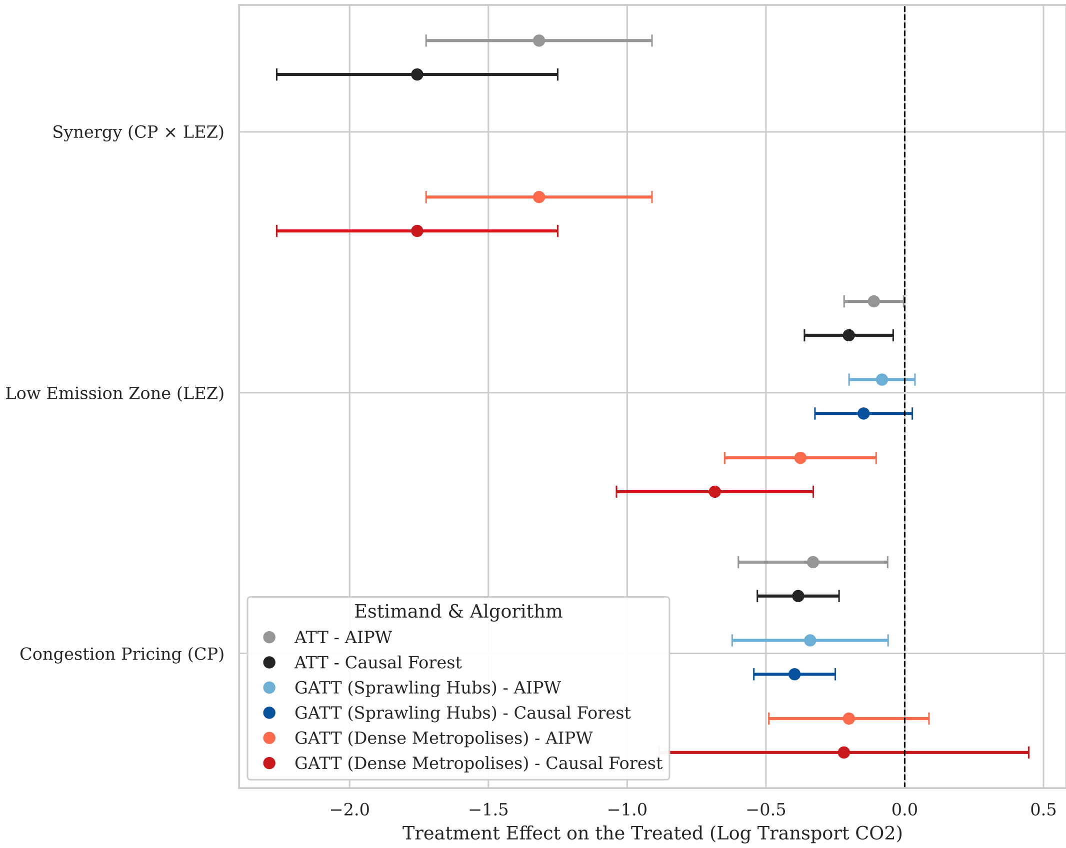

Figure 3: Policy Efficacy by Algorithm (95% CI)

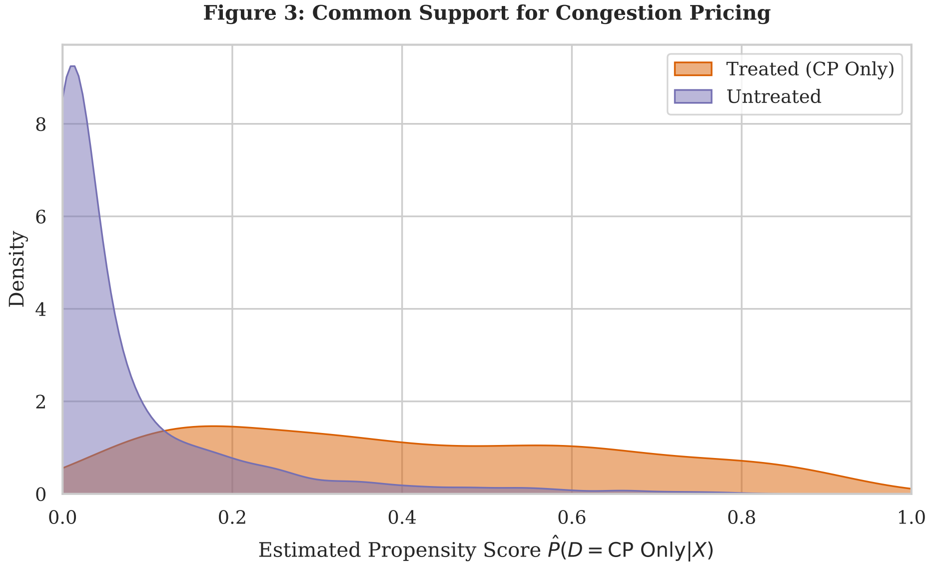

Figure 4: Common Support for Congestion Pricing

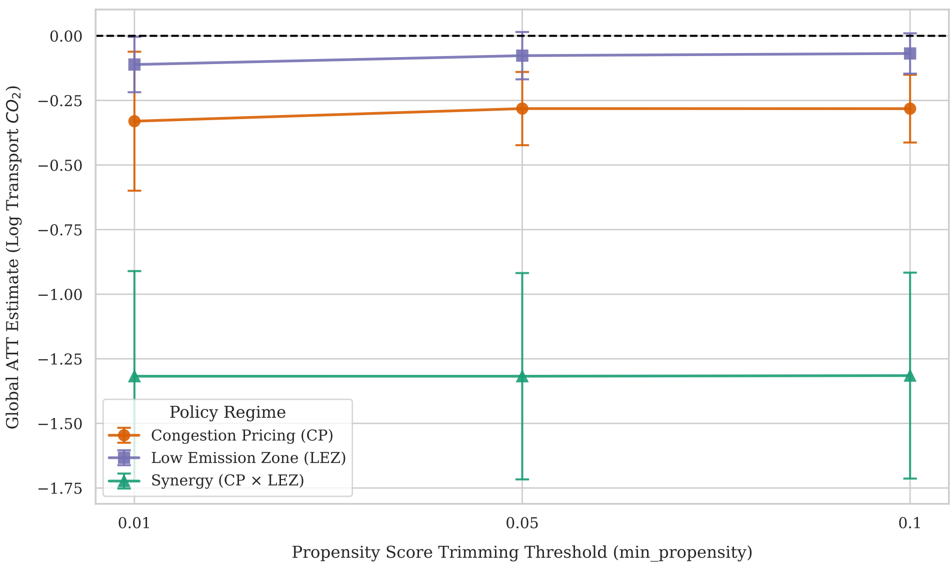

Figure 5: Sensitivity to Positivity Trimming

Declaration of Autorship

I confirm that this report is based on my own work. In preparing this report, I have not received any help from another human nor have I discussed any aspects of the empirical project with others.

Ferdinand Loser